

Esercizio

$$\begin{cases} \dot{x}_1(t) = x_2(t) + x_1(t) u(t) \\ \dot{x}_2(t) = -x_1(t) + x_2^2(t) \end{cases}$$

Equilibri $x_1(t) = \bar{x}_1$, $x_2(t) = \bar{x}_2$, $u(t) = \bar{u}$

$$\begin{cases} 0 = \bar{x}_2 + \bar{x}_1 \bar{u} \\ 0 = -\bar{x}_1 + \bar{x}_2^2 \end{cases} \quad 2 \text{ eq. } 3 \text{ incognite}$$

- Fisso $\bar{u} = 1$, trovare gli equilibri se esistono

$$\begin{cases} 0 = \bar{x}_2 + \bar{x}_1 & \rightarrow & \bar{x}_2 = -\bar{x}_1 & \bar{x}_2 = 0 & \bar{x}_2 = -1 \\ 0 = -\bar{x}_1 + \bar{x}_2 & & -\bar{x}_1 + \bar{x}_1^2 = 0 & \bar{x}_1 = 0, & \bar{x}_1 = 1 \end{cases}$$

Eq. 1: $(0, 0, 1)$

Eq. 2: $(1, -1, 1)$

- Fisso $\bar{u} = 4$ trovare gli equilibri se esistono:

$$\begin{cases} 0 = \bar{x}_2 + 4\bar{x}_1 & \bar{x}_1 = -\frac{\bar{x}_2}{4} = \begin{cases} -1/2 \\ +1/2 \end{cases} \\ 0 = -4 + \bar{x}_2^2 & \rightarrow & \bar{x}_2 = \pm 2 \end{cases}$$

Eq. 3: $(4, 2, -1/2)$

Eq. 4: $(4, -2, 1/2)$

• Fisso $\bar{x}_1 = -4$, trovare gli equilibri se esistono:

$$\begin{cases} 0 = \bar{x}_2 - 4\bar{x}_1 \\ 0 = 4 + \bar{x}_2^2 \leftarrow \text{MAI (sempre falso)} \end{cases}$$

■ Linearizziamo nell'intorno dell'Eq. 2: $(1, -1, 1)$

$$\begin{cases} \dot{x}_1 = x_2 + x_1 \mu_1 = f_1(x_1, x_2, \mu_1) \\ \dot{x}_2 = -x_1 + x_2^2 = f_2(x_1, x_2, \mu_1) \end{cases}$$

$$\tilde{x}_1(t) \stackrel{\text{def.}}{=} x_1(t) - \bar{x}_1, \quad \tilde{x}_2(t) \stackrel{\text{def.}}{=} x_2(t) - \bar{x}_2, \quad \tilde{\mu}_1(t) \stackrel{\text{def.}}{=} \mu_1(t) - \bar{\mu}_1$$

$$\left. \frac{\partial f_1}{\partial x_1} = \mu_1 \right|_{\bar{x}, \bar{\mu}} = 1 \quad \left. \frac{\partial f_1}{\partial x_2} = 1 \right|_{\bar{x}, \bar{\mu}} = 1 \quad \left. \frac{\partial f_1}{\partial \mu_1} = x_1 \right|_{\bar{x}, \bar{\mu}} = 1$$

$$\left. \frac{\partial f_2}{\partial x_1} = -1 \right|_{\bar{x}, \bar{\mu}} = -1 \quad \left. \frac{\partial f_2}{\partial x_2} = 2x_2 \right|_{\bar{x}, \bar{\mu}} = -2 \quad \left. \frac{\partial f_2}{\partial \mu_1} = 0 \right|_{\bar{x}, \bar{\mu}} = 0$$

$$\dot{\tilde{x}}(t) = A \tilde{x}(t) + B \tilde{\mu}(t) \quad A = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$